

torch.diag_embed#

https://pytorch.org/docs/stable/generated/torch.diag_embed.html

Description:

Creates a tensor whose diagonals of certain 2D planes (specified by dim1 and dim2) are filled by input. To facilitate creating batched diagonal matrices, the 2D planes formed by the last two dimensions of the returned tensor are chosen by default. The argument offset controls which diagonal to consider: If offset = 0, it is the main diagonal. If offset > 0, it is above the main diagonal. If offset < 0, it is below the main diagonal.

Function Signature

```
torch.diag_embed(input, offset=0, dim1=-2, dim2=-1) →  
Tensor#
```

Parameters

Code Examples

```
>>> a = torch.randn(2, 3)
>>> torch.diag_embed(a)
tensor([[[ 1.5410,  0.0000,  0.0000],
          [ 0.0000, -0.2934,  0.0000],
          [ 0.0000,  0.0000, -2.1788]],
        [[ 0.5684,  0.0000,  0.0000],
          [ 0.0000, -1.0845,  0.0000],
          [ 0.0000,  0.0000, -1.3986]]])

>>> torch.diag_embed(a, offset=1, dim1=0, dim2=2)
tensor([[[ 0.0000,  1.5410,  0.0000,  0.0000],
          [ 0.0000,  0.5684,  0.0000,  0.0000]],
        [[ 0.0000,  0.0000, -0.2934,  0.0000],
          [ 0.0000,  0.0000, -1.0845,  0.0000]],
        [[ 0.0000,  0.0000,  0.0000, -2.1788],
          [ 0.0000,  0.0000,  0.0000, -1.3986]],
        [[ 0.0000,  0.0000,  0.0000,  0.0000],
          [ 0.0000,  0.0000,  0.0000,  0.0000]]])
```

Detailed Documentation

Documentation

Creates a tensor whose diagonals of certain 2D planes (specified by dim1 and dim2)

are filled by input. To facilitate creating batched diagonal matrices, the 2D planes formed by the last two dimensions of the returned tensor are chosen by default.

The argument `offset` controls which diagonal to consider:

If `offset = 0`, it is the main diagonal.

If `offset > 0`, it is above the main diagonal.

If `offset < 0`, it is below the main diagonal.

The size of the new matrix will be calculated to make the specified diagonal of the size of the last input dimension. Note that for `offset` other than 000, the order of `dim1` and `dim2` matters. Exchanging them is equivalent to changing the sign of `offset`.

Applying `torch.diagonal()` to the output of this function with the same arguments yields a matrix identical to input. However, `torch.diagonal()` has different default dimensions, so those need to be explicitly specified.

`input (Tensor)` – the input tensor. Must be at least 1-dimensional.

`offset (int, optional)` – which diagonal to consider. Default: 0 (main diagonal).

`dim1 (int, optional)` – first dimension with respect to which to take diagonal. Default: -2.

`dim2 (int, optional)` – second dimension with respect to which to take diagonal. Default: -1.